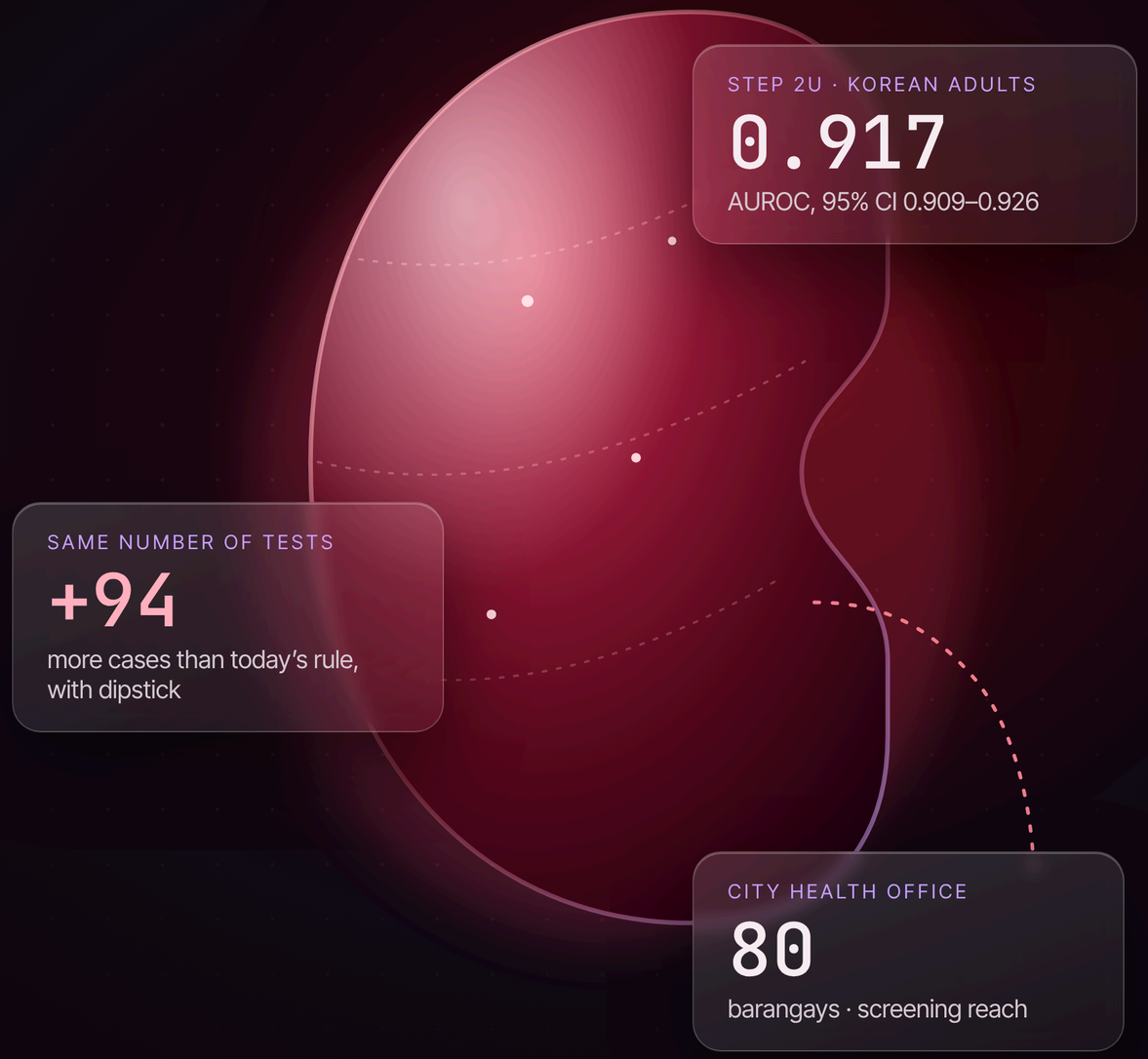
NEPHLUX

Who should get the kidney test *first?*

Machine learning–based albuminuria risk stratification from PhilHealth YAKAP labs, for early kidney damage.

Resident check → YAKAP labs + dipstick → City Health Office

PJDSC 2026 · Public Health Analytics · Set in Cagayan de Oro City



NEPHLUX supports screening; it does not diagnose. Tested on U.S. and Korean adults · Philippine testing is the next step.

Dialysis is PhilHealth's top-paid procedure.

Hemodialysis led PhilHealth's medical procedure payments in fiscal year 2024 (PhilHealth, 2025).

₱990,600

a year for one patient: **₱6,350** a session × **156** sessions (PhilHealth, 2024b)

<10%

of Filipino adults had been diagnosed with CKD in 2021 (Orejas, 2024)

Stock photo: Bas Peperzak, Cebu City (Unsplash). Not a patient.

In Cagayan de Oro, kidney disease arrives at stage 5.

One year of adult admissions at Northern Mindanao Medical Center, May 2024 to May 2025.

1,207
CKD admissions, of 32,378 adults

580
from Cagayan de Oro City, the largest share of any city or town

82.4%
arrived at **stage 5**

98.4%
were **already on dialysis**

Leading causes: high blood pressure **35.7%**, diabetes **34.8%**.

Hospital admissions, not a city rate (Mambering & Espiga-Comonsad, 2026). No published source gives CKD by Philippine city or barangay.

YAKAP pays for the labs, but not the test that finds early damage.

YAKAP primary care lab list (PhilHealth, 2024a)

- HbA1c
- Creatinine
- Lipid profile
- CBC with platelets
- Urinalysis (dipstick)
- FBS
- OGTT
- ECG
- Chest X-ray
- Sputum
- Ultrasound
- Mammogram
- Pap smear
- Fecalysis
- FOBT
- uACR · not covered

Highlighted: the YAKAP results NEPHLUX reads. The urine albumin-creatinine ratio (uACR) is the KDIGO test for early kidney damage (KDIGO, 2024).

85%

of adults with uACR ≥30 mg/g had a **normal-looking creatinine** (eGFR ≥60) in U.S. survey data (CDC, n.d.)

24%

had **neither diabetes nor high blood pressure**, so the usual rule never sends them

A YAKAP clinic cannot send everyone for a uACR test, and has no evidence-based way to choose who goes first.

PHOTO OF INTERVIEWEE
PERSON LIVING WITH A KIDNEY-RELATED PROBLEM

INTERVIEW · PERSON LIVING WITH A KIDNEY-RELATED PROBLEM

KEN L. PEÑAFLORIDA

A resident diagnosed with a kidney-related problem

Before the diagnosis

> *"Nagpa-check up lang ako for my BP and sugar. Normal naman daw ang iba, kaya akala ko okay na ako. Walang nagsabi sa akin na puwede palang tingnan ang kidney."*

What I didn't know

> *"Hindi ko alam na may YAKAP clinic pala, at covered ang ilang lab tests. Kung alam ko, nagpa-check sana ako bago pa lumala."*

What I wish had happened

> *"Sana may nagsabi sa akin: 'Ito ang risk mo, magpa-urine test ka na.' Hindi ko kailangan ng bagong test, kailangan ko lang malaman na dapat."*

Rank who gets the uACR test, from results the clinic already has.

The nurse or doctor enters the labs. The doctor decides.

1

Resident check

10 questions on a phone, no labs. Awareness, plus a clinic code.

2

YAKAP check-up

BP, HbA1c, creatinine, lipids, CBC and the urine dipstick YAKAP already covers.

3

Step 2U model

Chance of uACR ≥30 mg/g and its top reasons. No dipstick? It uses Step 2, labs only.

4

Doctor decides

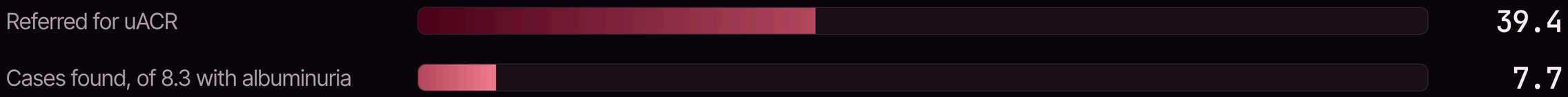
Approve, override with a reason, or no referral. Dipstick protein 1+ always means refer.

5

uACR test

The one test YAKAP does not cover. The clinic follows up its own referrals.

Per 100 adults in the Korean test set, with the same referrals as today's rule



We wrote the pass mark before we trained anything.

Data

Korea's national health survey (KNHANES; KDCA, n.d.)

It measures the dipstick **and** uACR on the same adults. The U.S. survey has no dipstick.

Learned on **2021–2024**: **20,465** adults.

Tested once on **2019–2020**: **11,047** adults, 912 cases.

Model

Penalised logistic regression

Inputs: age, sex, history, BMI, measured BP, HbA1c, eGFR (CKD-EPI 2021; Inker et al., 2021), cholesterol, blood count, and dipstick protein, blood, glucose and specific gravity.

Simple on purpose: every reason can be shown to the doctor.

Pass mark, written first

Fixed before the test years were opened

1. AUROC 95% CI above 0.70.

2. More cases than "diabetes, high BP or dipstick 1+" at the same number of referrals.

3. Better than the same model without the dipstick.

4. Calibration slope 0.8–1.2.

Script and model hashed (SHA-256) before the one test run. 1,000 bootstrap resamples. Reported with TRIPOD+AI (Collins et al., 2024). Adults 20+, not pregnant, no known kidney disease.

STEP 2U · KOREAN TEST SET · SCORED ONCE

+94

more cases found than referring everyone with diabetes, high BP or dipstick protein 1+, with the same number of uACR tests.

Of 912 cases among 11,047 Korean adults (KNHANES 2019–2020) · 95% CI +72 to +118 · 39.4% referred by both · sensitivity 92.8% vs 82.5%

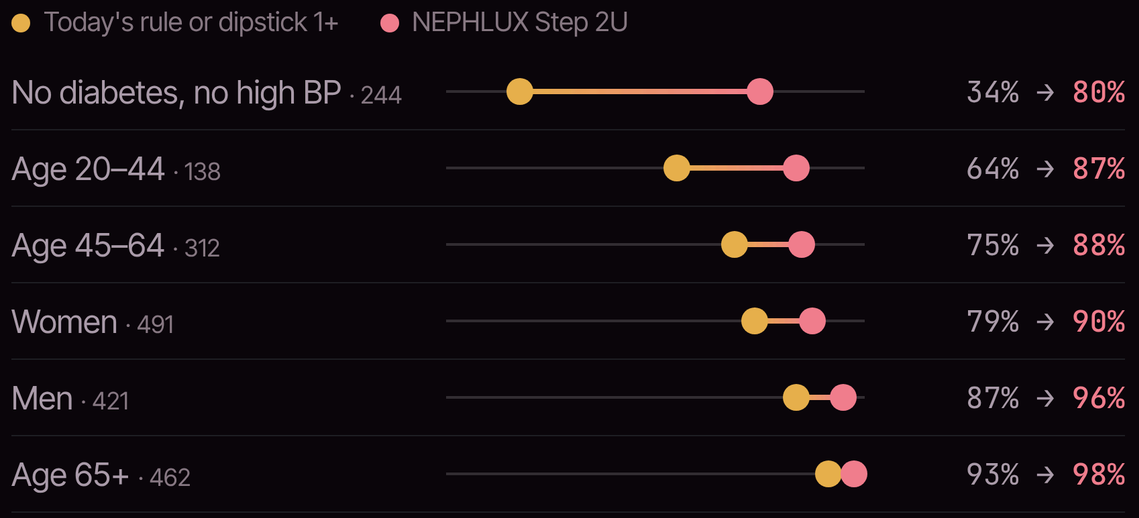
The biggest gain is in people today's rule never sends.

No diabetes and no high blood pressure · 244 cases

80% caught by **NEPHLUX**

34% by today's rule

More than 1 in 4 cases in the test set had neither condition. The dipstick and the labs find them; a yes/no history question cannot.



Share of cases caught, same number of referrals (39.4%)

KNHANES 2019–2020 test set, 11,047 Korean adults, 912 cases.

When tests are scarce, NEPHLUX finds four times the cases.

With 50 uACR tests per 1,000 adults

NEPHLUX order

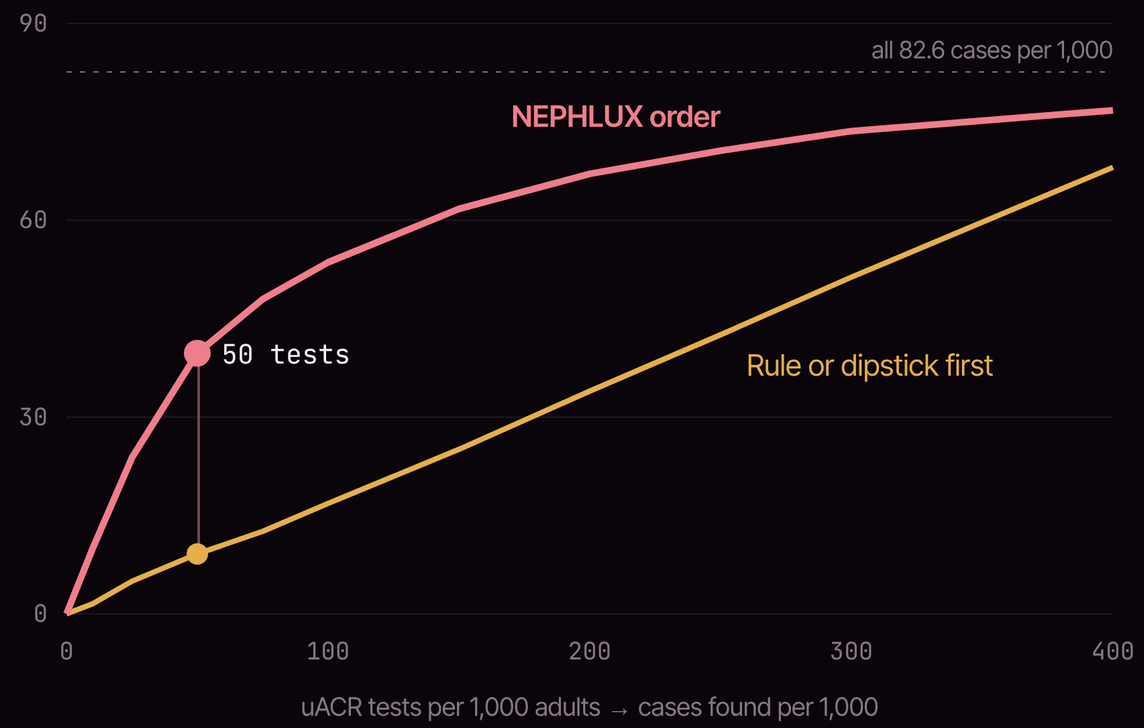
39.7

cases found · about **₱755** per case

Today's rule or dipstick first

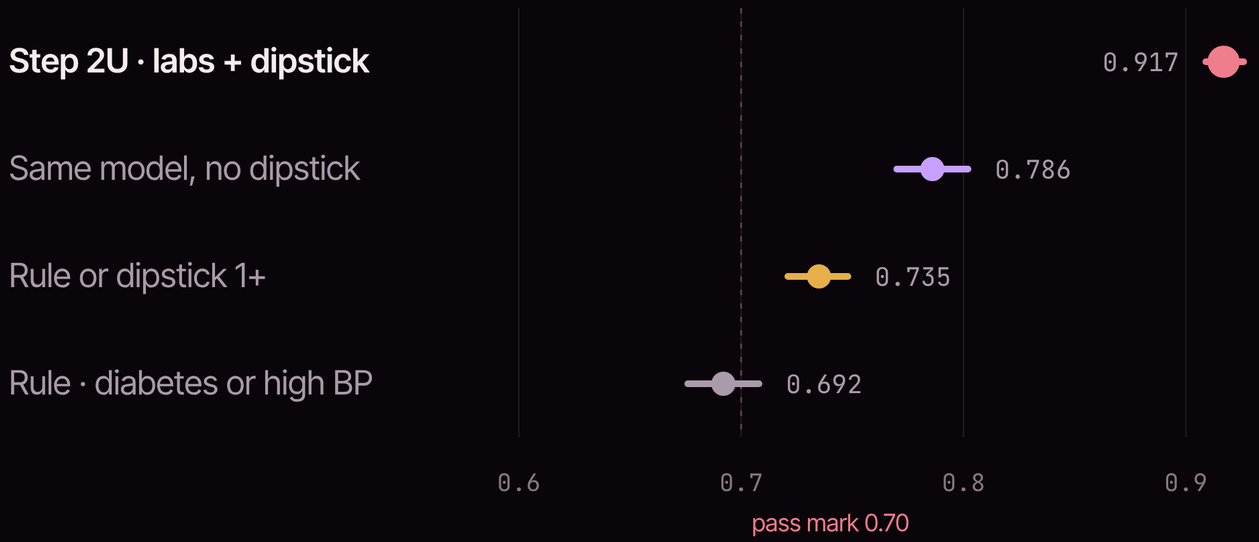
9.1

cases found · about ₱3,281 per case



KNHANES 2019–2020 test set. uACR price assumed at ₱600 for a public or bulk purchase (range ₱300–1,200). Published retail price: ₱1,500–3,500 (Baclig, 2026), which gives ₱1,887–4,404 vs ₱8,203–19,141 per case. Cases found, not dialysis avoided.

It ranks better than every simple rule, and its percentages hold.



AUROC, 95% bootstrap CI. Higher means better ranking of who has albuminuria.

KNHANES 2019–2020, 11,047 Korean adults, frozen model, scored once. All four pass-mark criteria met.

0.917

AUROC (95% CI 0.909–0.926). Survey-weighted: 0.927.

1.02

calibration slope (0.96–1.07; Van Calster et al., 2019): a **10%** estimate means about 10%. Mean predicted 8.5%, observed 8.3%.

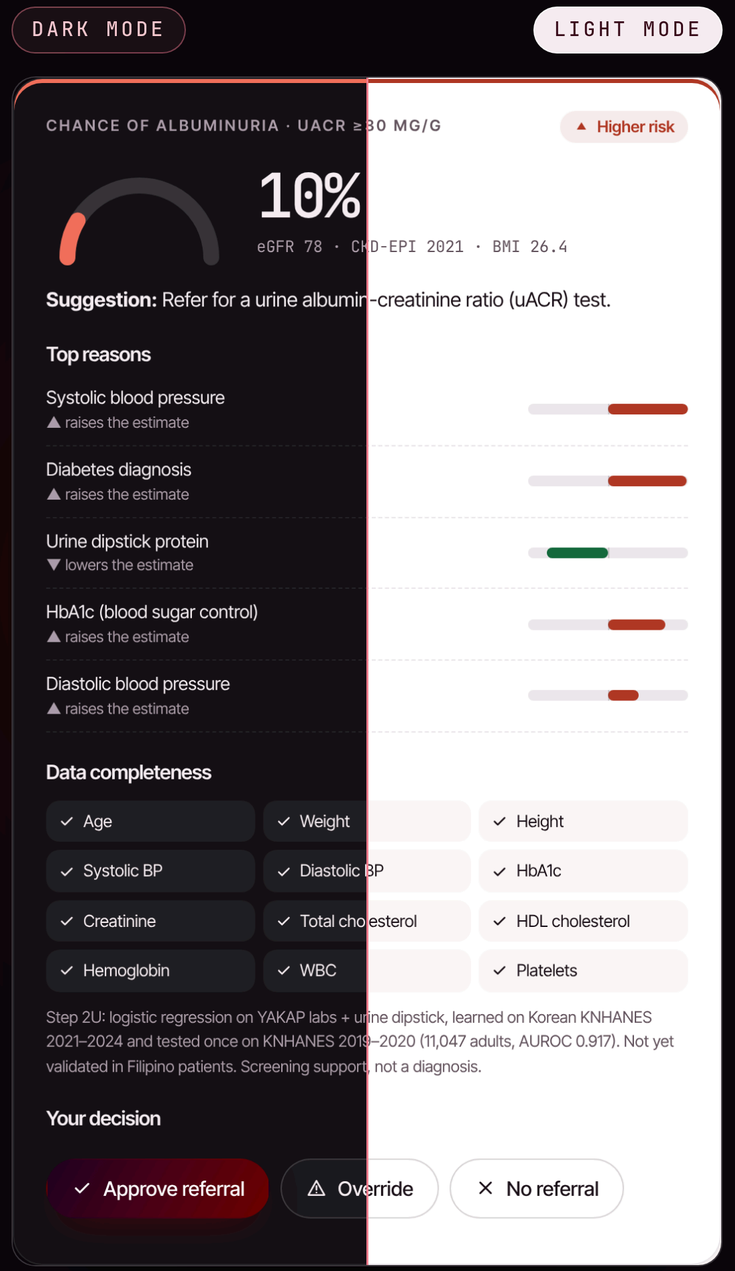
+0.131

AUROC added by the YAKAP dipstick (0.116–0.147).

The model suggests. The doctor decides.

- A nurse or doctor enters **labs, BP and the YAKAP dipstick** in one form.
- A chance and its **top reasons**, nothing hidden.
- A missing lab **stops the estimate**; nothing is guessed.
- Approve, **override with a reason**, or decline.
- Prints the uACR referral slip.

Mang Ramon, 58, fictional. Dipstick negative, creatinine normal-looking (eGFR 78). Step 2U: **about 10%**, well above the 2.55% cut. Refer.



One web app, three people, one job each.

Resident

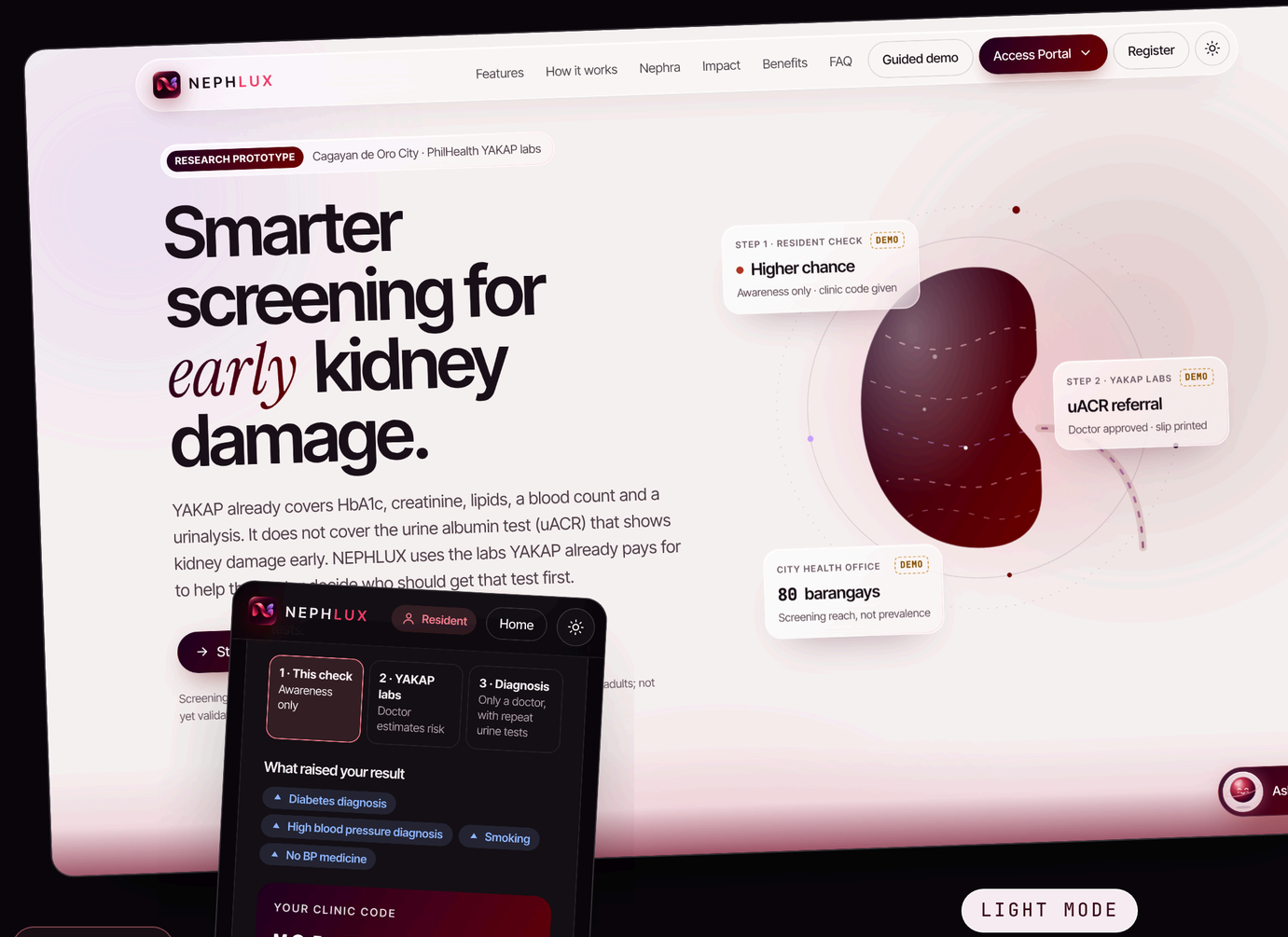
10 questions in English, Bisaya or Tagalog. A plain result, a 9-character clinic code and the nearest of 42 YAKAP clinics (PhilHealth, 2026). Nothing is saved.

YAKAP clinic, nurse or doctor

Loads the code and enters the labs and dipstick. The doctor decides and prints the slip. The clinic follows up referrals with no result after 14 days.

City Health Office

Anonymous counts for 80 barangays, a test budget slider and outreach priorities.



Stock photo: Rainier Ridao, Cagayan de Oro City (Unsplash). Not a patient.

From a phone at home to a referral slip.

DARK MODE

NEPHLUX · Resident · Home

RESIDENT CHECK · 2 MINUTES · NOTHING IS SAVED

Should I get my *kidneys* checked?

An awareness check from 10 questions, with no lab tests. It is not a diagnosis. Your answers stay on this screen; at the end you get a short code that carries them to a YAKAP clinic.

English | Bisaya | Tagalog

Bisaya and Tagalog are draft translations for review by native speakers.

QUESTION 10 OF 10

Imong barangay sa Cagayan de Oro

Your barangay in Cagayan de Oro

Bonbon

Resident. Two minutes, no account. A "higher" or "lower" chance, never a diagnosis.

LIGHT MODE

PROTOTYPE · SAMPLE RECORD · NOT FOR CLINICAL USE

Referral slip: urine albumin-creatinine ratio (uACR) test

Date 2026-10-02 · Seen at City Health Office (CHO) (public YAKAP clinic), Cagayan de Oro City

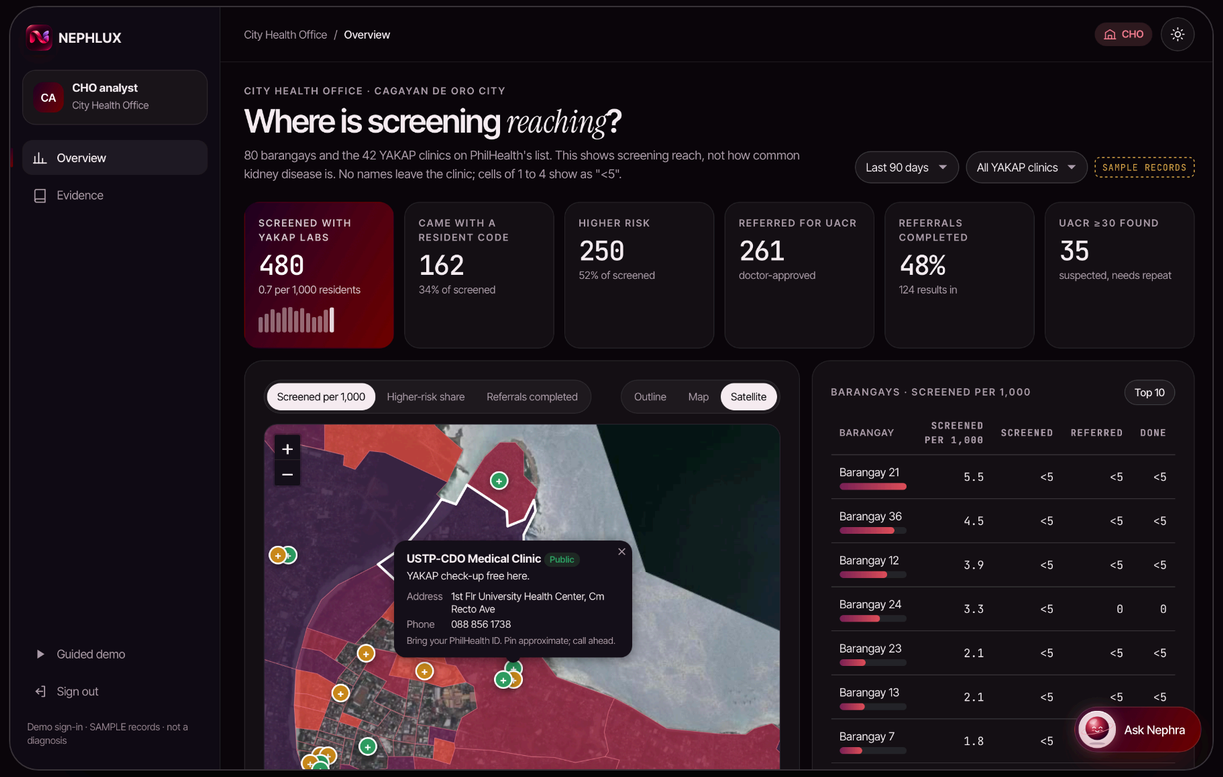
Patient name	_____ (write by hand; the app keeps initials only)
Patient	R.S., 58 y, M, Bonbon, Cagayan de Oro City
Resident code	MSJ-5WP-N1Y (answers confirmed with the patient)
BP	148/88 mmHg
HbA1c	7.4%
Creatinine / eGFR	1.10 mg/dL / 78 mL/min/1.73m² (CKD-EPI 2021)
Urine dipstick	protein negative · blood negative · glucose trace · SG 1.020
Screening estimate	10% estimated chance of uACR ≥30 mg/g (higher risk; Step 2U, labs + dipstick)

Main reasons: Systolic blood pressure ↑ · Diabetes diagnosis ↑ · Urine dipstick protein ↓ · HbA1c (blood sugar control) ↑
Request: spot uACR, first-morning sample preferred. A single result ≥30 mg/g is **suspected albuminuria**, not a diagnosis:

Clinic. The slip carries the estimate, the reasons and the repeat-test rule. The doctor signs it.

The clinic code **is** the resident's answers packed into 9 characters with a check value, so nothing is typed twice and nothing is stored.

Every referral becomes local evidence the city does not have yet.



Today, in the app

Each clinic sends the City Health Office **counts only**: screened, higher risk, referred, uACR done, uACR raised, for all 80 barangays. Cells of 1 to 4 are hidden.

Next, with consent

Each returned uACR is a Filipino record with **labs, dipstick and uACR together**: the data needed to test NEPHLUX on Filipinos, and a start toward a Philippine kidney dataset held by the city.

Consent under the Data Privacy Act of 2012 · the CHO as data controller · research extracts only with ethics approval. Clinic visitors, not a survey sample: it shows screening reach, not prevalence.

Dashboard records are SAMPLE data. Clinic pins are approximate. Populations: 2020 census (PSA, 2021; 728,402 residents).

Safe by design, with Nephra to help.

Privacy

- The resident check **stores nothing**: the server scores the answers and keeps no record. No account needed.
- The city sees **counts, not names**. The clinic that ordered the test follows it up.

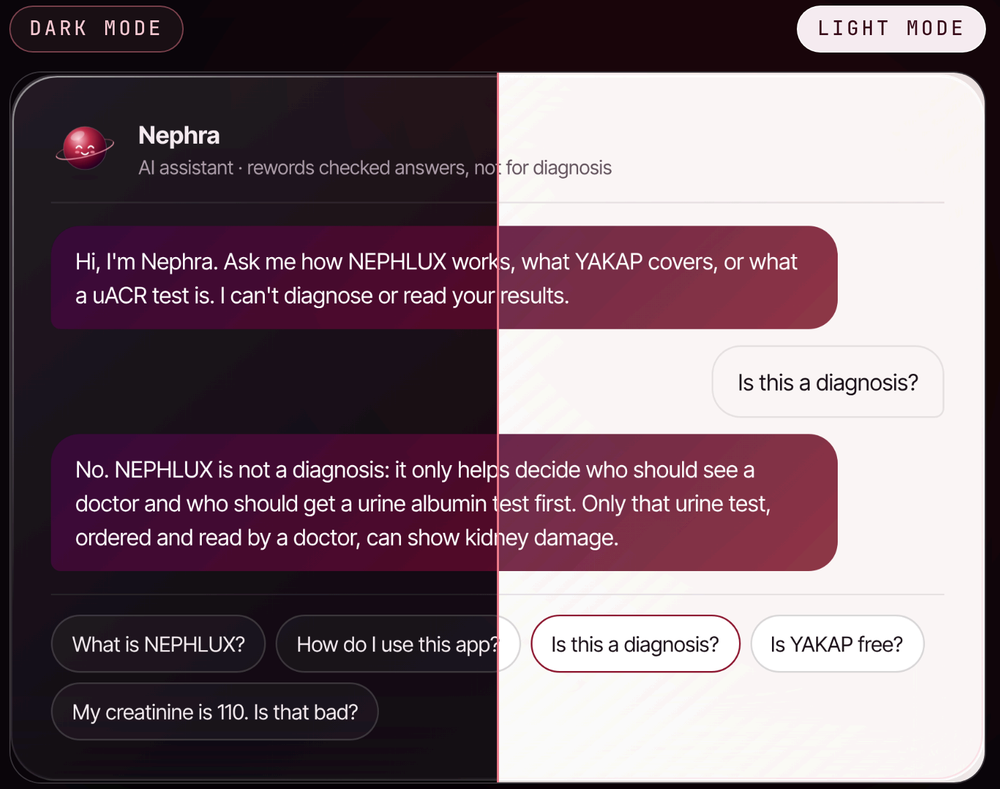
Clinical safety

- The doctor decides **every** referral; an override needs a written reason.
- Dipstick protein 1+ means refer, **whatever the score**. eGFR under 60 raises a warning.
- The app's risk engine **equals the Python model** on 300 test patients, checked by an automated test.

HAPPY

EXPLAINING

CONCERNED

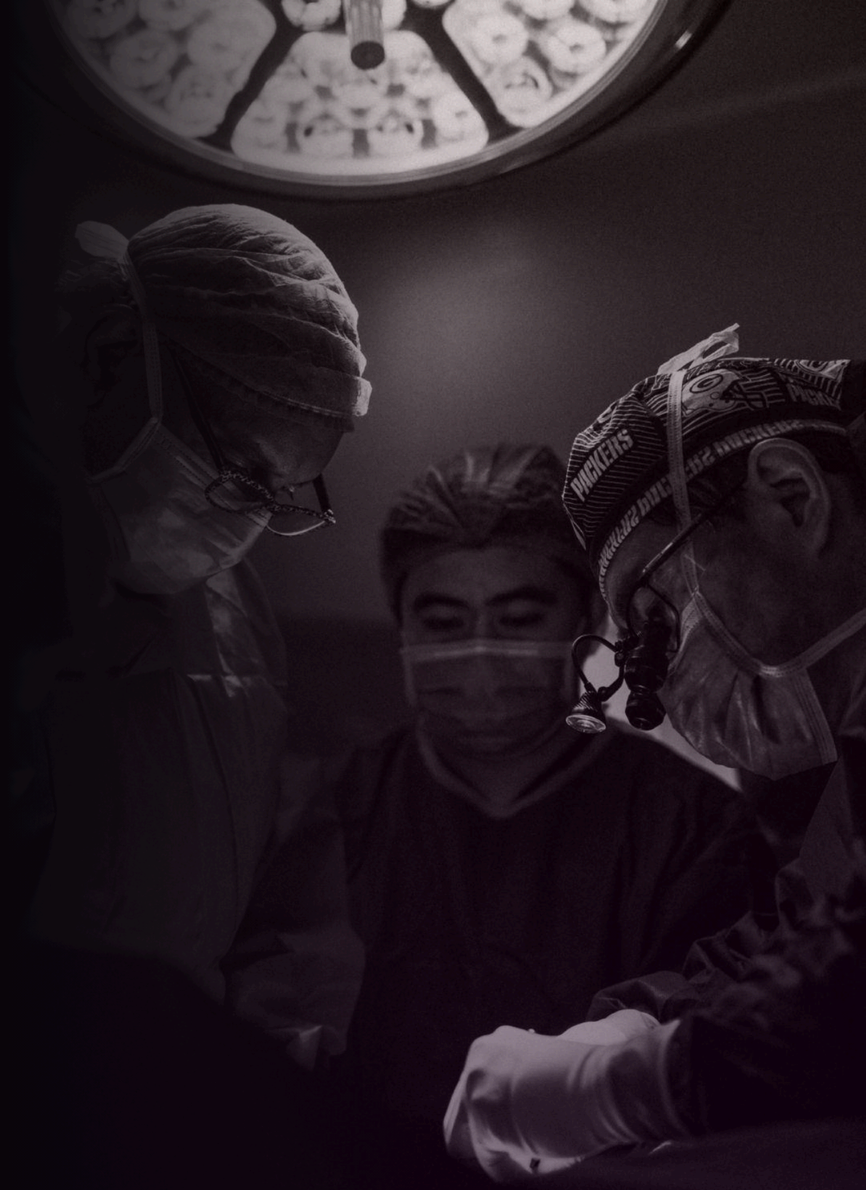


Nephra, the Orb, answers from **78 approved answers**. An AI model may reword them, but it cannot add facts, calculate or diagnose. The answer bank gave **0** unsafe answers on new test questions; danger signs always get the fixed emergency answer.

Next: test it on Filipino patients in Cagayan de Oro.

Tested once on Korean adults. The next test set is ours.

1. **Clinical review** by a YAKAP physician and a nephrologist.
2. **Walkthrough** with one YAKAP nurse or doctor on sample data; fix what they find. A usability test across several clinics comes later.
3. **Cagayan de Oro validation:** YAKAP records with uACR from about 2,800–4,400 adults (Riley et al., 2021), with ethics approval and a data-sharing agreement. Recalibrate, then score the frozen model once.
4. **Later:** a City Health Office pilot with a fixed uACR budget and a few barangays (needs their agreement).



Stock photo: JC Gellidon, Manila (Unsplash).

The labs are already paid for. NEPHLUX decides who gets the urine test first.

+94 cases at the same number of tests. AUROC 0.917, pre-registered and scored once on 11,047 Korean adults. Its next test is a Filipino patient.

NEPHLUX ranks who needs the urine test; the doctor decides and the uACR confirms. A working web app (React, Tailwind CSS, Zustand; NestJS and PostgreSQL API). Every figure regenerates from one script. Independent project, not affiliated with PhilHealth.

Stock photo: Jeremy Brady (Unsplash). Not a patient.

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